MummyAI

Maternal, Child Health & Development Companion

Project Proposal

Prepared for: Maternal and child health digital support initiative
Project type: Early-stage prototype and product proposal
Status: Concept-to-prototype development

Executive Summary

MummyAI is a privacy-first digital health companion designed to help families organize pregnancy and child-health information, understand approved educational content, track appointments and developmental observations, and prepare for informed conversations with healthcare professionals. The platform envisions a single accessible system for pregnancy, birth, infancy, and childhood care that reduces information fragmentation while supporting safe, understandable guidance.

This project is an early-stage local prototype built with a React frontend, an Express backend, and a restricted local AI agent. It supports general health education, clinic-question preparation, and synthetic record access in a tightly controlled environment. The project intentionally does not present itself as a medical device or clinical decision-support system and explicitly avoids real patient data.

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1. Background and Problem

Families often manage maternal and child health information across paper records, notes, mobile apps, hospital visits, and informal reminders. During pregnancy and early childhood, health information is fragmented and scattered across growth measurements, immunization records, feeding notes, appointments, developmental observations, and clinic conversations.

This fragmentation creates several challenges: difficulty preparing for appointments, missed follow-ups, limited understanding of health information, inconsistent record-keeping, and poor communication between caregivers and healthcare professionals. In multilingual or low-connectivity settings, families may struggle to interpret guidance or maintain continuity of care.

MummyAI responds to this need by creating a digital companion that supports healthcare information organization, simple-language explanations, reminders, and guided health conversations while operating under strict safety, privacy, and clinical governance rules.

2. Need and Opportunity

? Families need a secure place to store and review pregnancy, birth, and child-health records.

? Pregnant mothers, parents, and caregivers often need reminders, milestone tracking, and structured summaries for healthcare visits.

? Health information is often more useful when it is accessible in plain language and in a preferred local language.

? A safe, AI-assisted companion can reduce friction in health information access without replacing clinical judgment.

? A responsibly designed product can improve continuity of care, preventive engagement, and household readiness for medical conversations.

3. Project Vision

The long-term vision is a single family-centered digital platform spanning pregnancy, birth, infancy, and childhood. It supports digital records, multilingual access, reminders, preventive care organization, and safe AI assistance grounded in approved educational sources and user-authorized record access.

The platform is designed to support families in understanding health information without making autonomous medical diagnoses or prescribing care. It clearly separates AI-generated guidance from verified records and keeps healthcare professionals responsible for clinical decisions and consequential workflows.

4. Target Users

? Pregnant mothers who need antenatal summaries, reminders, and pregnancy education.

? Parents and caregivers tracking child growth, immunization, feeding, developmental milestones, and clinic attendance.

? Public health midwives and community health workers who need authorized follow-up workflows.

? Doctors and specialists who may require concise, consent-based summaries and referral context.

? Program administrators and health organizations focusing on preventive care and privacy-preserving reporting.

5. Product Scope

The project scope includes a digital record system, AI assistant, reminder functionality, and a safe educational interface. The MVP focuses on foundational capabilities that are usable, safe, and extensible.

? Pregnancy and maternal health information management.

? Birth and newborn record tracking.
? Child growth, immunizations, and appointments.
? Developmental observations, parent notes, and learning activities.
? General health education and clinic-visit preparation.
? Restricted AI tool use for bounded tasks and synthetic demo access.
? Professional review workflows for clinically consequential actions.

6. Core Features

? Pregnancy profile management and appointment tracking.
? Birth and newborn summary records with health history.
? Child health records including immunization and growth monitoring.
? Milestone tracking, sleep, feeding, and developmental notes.
? Educational AI chatbot for general questions and explanation.
? Clinic-visit preparation with structured questions and summaries.
? Reminder engine for appointments and preventive care.
? Multilingual interface support with future voice and low-connectivity features.

7. Functional Requirements

? User profile and caregiver relationship management.
? Validated pregnancy, birth, and child health record forms.
? Growth measurement tracking with unit validation and charting support.
? Appointment scheduling and reminder workflows.
? Immunization history and preventive-care reminders.
? Developmental observation capturing and timeline views.
? Chatbot interface for general guidance and clinic preparation.
? Restricted tool-calling layer with backend-enforced permissions.
? Synthetic data support for demos and evaluation use cases.

8. Non-Functional Requirements

? Privacy by design and least-privilege access.
? Role-based authorization and caregiver consent.
? Transparent data provenance for records and AI-generated content.
? Secure handling of sensitive data and audit preparation.
? Multilingual and low-connectivity friendly design.
? High reliability, maintainability, and clear separation between trusted logic and AI components.

9. Proposed Architecture

The architecture is intentionally simple and safety-oriented. The frontend is a React-based web experience that supports parents and future professional users. The backend is an Express API responsible for request validation, access control, and domain logic. The AI layer is separate from trusted application logic and can only invoke a narrow set of permitted backend tools.
In the current prototype, the AI agent uses a local Ollama service and limited tool-calling support. It can explain capabilities, prepare clinic questions, and read synthetic sample records only when explicitly configured. Database access is constrained to synthetic records only; the project does not permit direct access to real patient data or unrestricted database commands.

10. Technology Stack

Frontend: React + Vite + responsive web UI
Backend: Node.js + Express + validation and middleware
AI: Local Ollama with constrained tool-calling support

Database: MongoDB Atlas for synthetic demo data; production database to be reviewed
Security: Helmet, rate limiting, CORS controls, and server-side validation
Planned support: RAG over approved sources, reminders, OCR, analytics, and audit workflow tools

11. AI System Design

The AI assistant is designed as a bounded tool-using agent rather than a freeform health advisor. It accepts user input, constructs a safety-scoped prompt, and decides whether a permitted tool should be called. Only a small allowlist of tools is exposed to the model, such as capability explanation, clinic-question preparation, and read-only synthetic data retrieval.
Every tool invocation is validated against a schema. Responses are filtered and labeled as either a verified record, general educational content, or AI-generated output. The system is designed to avoid making unsupported medical claims or silently turning AI outputs into verified record data.

12. Safety and Clinical Governance

MummyAI is intended to support education and organization, not replace healthcare professionals. It should be governed by strict clinical and ethical boundaries at every stage of development.

? Never diagnose from a single measurement, image, or conversation.
? Never recommend medication changes or treatment decisions without clinician oversight.
? Never delay urgent care with unnecessary follow-up questions.
? Never silently convert an AI extraction into a verified medical fact.
? Never expose one family?s records to another user.
? Always separate educational advice from verified records.
? Require consent and authorization before accessing or sharing records.

13. Privacy, Security, and Compliance

? Apply least-privilege access for caregivers, midwives, doctors, administrators, and support staff.
? Keep data collection minimal and necessary only for health support.
? Maintain audit trails for record access and AI tool use.
? Support secure retention, delete workflows, backups, and incident response.
? Restrict the prototype to synthetic records until privacy and authorization controls are reviewed.
? Assess legal, privacy, and health-data obligations relevant to the target country and deployment environment.

14. Risks and Mitigations

? Clinical safety risk: mitigate with professional review, approved content, and careful uncertainty language.
? Data privacy risk: mitigate using access controls, consent workflows, and audit logging.
? AI hallucination risk: mitigate with source grounding, validation, and explicit provenance labeling.
? Adoption risk: mitigate through accessible design, simple workflows, and multilingual support.
? Operational risk: mitigate through iterative testing, phased rollout, and governance reviews.

15. Development Roadmap

? Phase 0: Starter scaffold complete with frontend, backend, documentation, and safety notes.
? Phase 1: Product foundation with profile systems, forms, role boundaries, and privacy controls.
? Phase 2: Trusted AI with restricted tool-calling and synthetic record access.
? Phase 3: Growth monitoring, immunization tracking, reminders, and documentation workflows.
? Phase 4: Pilot evaluation, privacy/security assessment, accessibility testing, and governance approval before real-world use.

16. Success Metrics

? Successful completion of core pregnancy and child health record workflows.
? User-friendly experience for parents and caregivers.
? Safe and understandable AI answers for educational and clinic-preparation tasks.
? Strong outcomes in privacy, security, and authorization testing.
? Positive feedback from pilot users and front-line health workers.

17. Business and Social Impact

MummyAI can create meaningful value by reducing the burden of fragmented record management and improving access to understandable information in pregnancy and early childhood. The system can support preventive care engagement, timely reminders, and more informed clinic visits.
At a social level, the project can help families navigate health information in accessible ways, especially in multilingual environments and settings where digital tools can supplement but not replace healthcare support. The product has the potential to improve continuity of care while respecting professional roles and safety boundaries.

18. Conclusion

MummyAI is a responsible and safety-oriented concept for a digital health companion designed around family needs and public-health goals. The project balances genuine user value with strict privacy, clinical, and ethical guardrails. By building a strong product vision, a constrained AI architecture, and a phased roadmap, MummyAI can progress from a prototype into an evaluated and regulated health-support platform only after professional and legal review.
With careful stewardship, MummyAI can become a useful tool for education, organization, and preventive care while keeping medical professionals responsible for diagnosis, treatment, and final clinical judgment.

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19. Current Prototype Snapshot

? React frontend with Vite for the user interface.
? Express API with health checks and safety-focused middleware.
? Local Ollama-based AI agent with restricted prototype tools.
? Optional MongoDB Atlas connection for synthetic sample records only.
? Safety notes, architecture guidance, and roadmap documentation included in the repo.

20. Final Recommendation

The project should proceed as a carefully scoped digital companion and safe prototype. The next phase should prioritize authentication, caregiver authorization, validated record handling, and clear user journeys. A strong evaluation process, including privacy review, safety testing, and usability assessment, is essential before any future clinical pilot or real data use.